

---

**Proposal (One-Pager)****PROJECT PROPOSAL: OPTIMIZING WAGON PROCUREMENT & FREIGHT CONTRACTING AT CIL (LAKHANPUR AREA)**

---

**Problem Statement**

We examine the outbound rail logistics bottleneck at Coal India Limited (CIL) – Mahanadi Coalfields Limited (MCL), Lakhanpur Area. Shift-wise coal production (~60,729 tonnes/day) and demand (~67,009 tonnes/day) exhibit significant variance, resulting in an average dispatch deficit of ~3,123 tonnes/shift due to railway rake availability constraints. Relying solely on spot Indian Railways (IR) rake allotments via FOIS exposes CIL to wagon shortages during demand surges and demurrage/overbooking charges during slumps. We answer: How can CIL mitigate rake supply risks by designing an optimal strategic mix of captive/leased wagon fleets (via IR's GPWIS/SFTO schemes) versus spot IR rake bookings, backed by Long Term Tariff Contracts (LTTC) for long-distance power plant corridors?

**Plan**

We build a decision-making framework to evaluate the optimal wagon strategy using a two-tier fleet approach (Base Fleet vs. Flexible Fleet). First, we analyze the shift-wise production, demand, and dispatch variability using empirical data from Lakhanpur OCP. Second, we construct a total cost-minimization model that weighs the fixed costs of captive/leased rakes (accounting for ~4,000-tonne rake capacity and GPWIS freight rebates) against the variable expenses of spot railway bookings (including demurrage/wharfage fees and penalties for unfulfilled dispatches). Finally, we perform scenario and sensitivity analyses across different demand regimes and key transport corridors.

**Data and Information**

- Primary Source: Shift-wise coal production, demand, and dispatch dataset from Lakhanpur OCP, MCL (908 shifts / 303 days).
- Secondary Sources: Indian Railways / FOIS freight tariff schedules, IRCTC Commercial Wing policy guidelines on GPWIS, SFTO, and LTTC schemes, wagon payload standards, and CIL annual logistics reports.

**Expected Outcome**

The deliverable is a decision framework and wagon contracting matrix recommending the optimal captive-to-spot wagon ratio and contract structure. Success will be evaluated by a quantified cost-risk comparison demonstrating reduced dispatch deficits and minimized demurrage penalties.